

Software Engineering

Chapter 1- (The Product)

Lecture 01

Farjana Parvin
Lecturer
Dept. of CSE, RUET

Reference Books

1. *Software Engineering, A practitioner's Approach (5th Edition)*, Roger S. Pressman
2. *Software Engineering*, Ian Sommerville
3. *Design Patterns, Elements of Reusable Object-Oriented Software*, Erich Gama, Richard Helm, Ralph Johnson, John Vlissides

Software (Product)

- Software is
 - Instructions (computer programs) that when executed provide desired function and performance,
 - Data structures that enable the programs to adequately manipulate information, and
 - Documents that describe the operation and use of programs.

Software characteristics

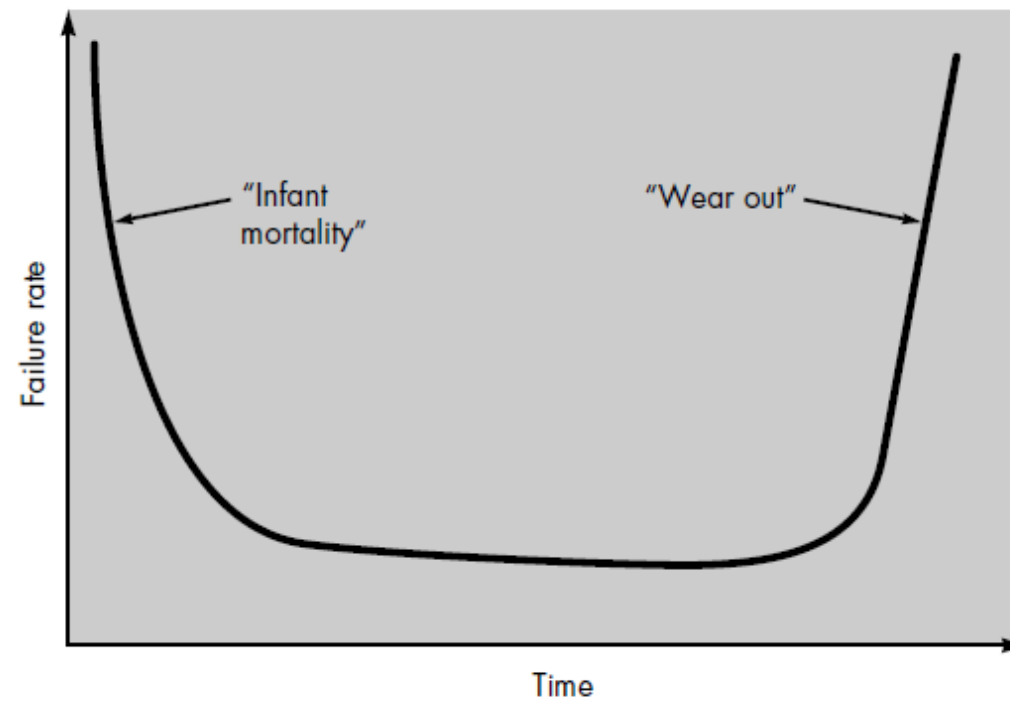
- **Software is developed or engineered, it is not manufactured in the classical sense.**
 - Two differences exists between software and hardware manufacture.
 - The manufacturing phase for hardware can introduce quality problems that are easily corrected for software.
 - Both activities require the construction of a “product”, but the approaches are different.

Software characteristics

- **Software doesn't “wear out” but it does deteriorate.**
 - Hardware failures are high early in its life, failure rate drops to a steady-state level for some period of time and the failure rate rises again as hardware components suffer from the cumulative affects of dust, vibration, etc.
 - For software, undiscovered defects caused high failure rates early in the life of a program, those were corrected without introducing other errors.

Software characteristics

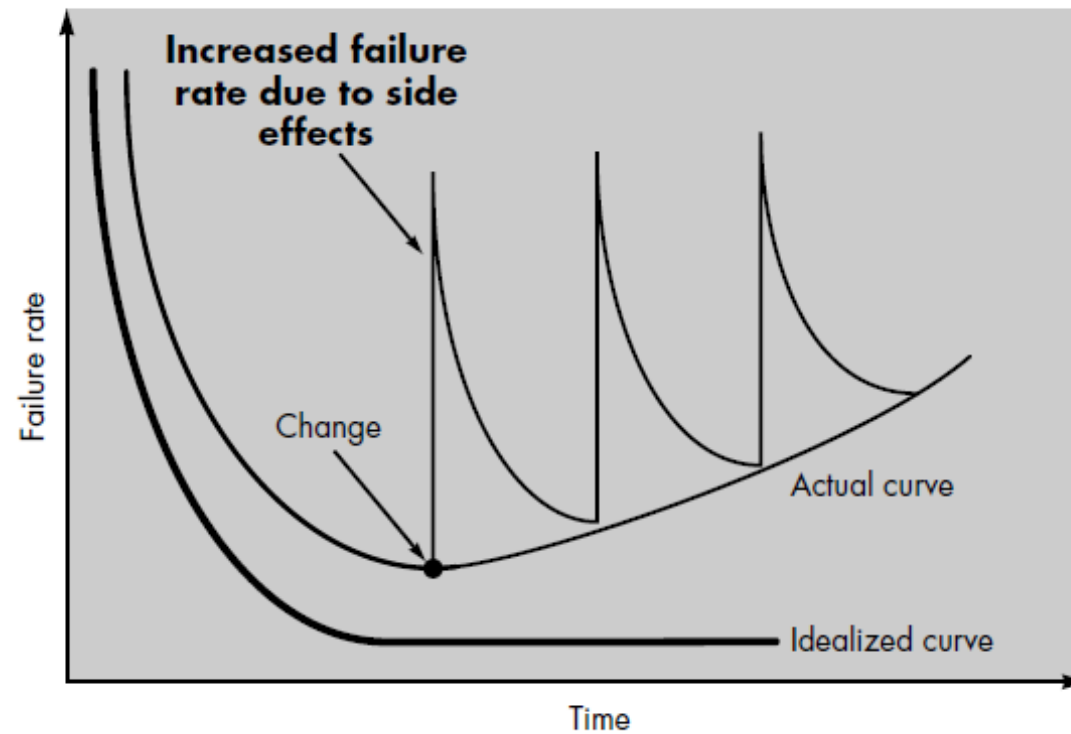
FIGURE 1.1
Failure curve
for hardware



Software characteristics

FIGURE 1.2

Idealized and actual failure curves for software



Software characteristics

- **Although the industry is moving toward component-based assembly, most software continues to be custom built.**
 - Hardware manufacture for a microprocessor-based product, the design engineer draws a simple schematic of the digital circuitry, does some fundamental analysis to ensure that proper function will be achieved, and then refers to the catalog of digital components,
 - No catalogs for the software components. It is possible to order off-the-shelf software, but only as a complete unit, not as components that can be reassembled into new programs.

Software applications

- **System software**

- System software is a collection of programs written to service other programs. System software has heavy interaction with computer hardware, heavy usage by multiple users, etc.

- **Real-time software.**

- Software that monitors/analyzes/controls real-world events as they occur is called real time.
 - Elements of real-time software include a data gathering component.

Software applications

- **Business software**

- Business information processing is the largest single software application area.
- E.g. payroll, accounts receivable, inventory, etc.

- **Engineering and scientific software.**

- Engineering and scientific software have been characterized by "number crunching" algorithms.
- E.g. Computer-aided design, system simulation, and other interactive applications.

Software applications

- **Embedded software.**

- Intelligent products have become commonplace in nearly every consumer and industrial market. Embedded software resides in read-only memory and is used to control products and systems for the consumer and industrial markets.
- Embedded software can perform very limited and esoteric functions (e.g., keypad control for a microwave oven)
- Provide significant function and control capability (e.g., digital functions in an automobile such as fuel control, dashboard displays, and braking systems)

Software applications

- **Personal computer software**

- Word processing, spreadsheets, computer graphics, multimedia, etc, are only a few of hundreds of applications.

- **Web-based software**

- The Web pages retrieved by a browser are software that
 - incorporates executable instructions (e.g., CGI, HTML, Perl, or Java), and data (e.g., hypertext and a variety of visual and audio formats).

Software applications

- **Artificial Intelligence software**

- Artificial intelligence software makes use of non numerical algorithms to solve complex problems that are not amenable to computation or straightforward analysis.
- E.g. AI software are pattern recognition, theorem proving and game playing.

Thank You